NLVAE: A New Machine Learning Approach for Extracting and Identifying Sales-Driving Product Attributes

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Motivation, Research Questions, and Contribution

Understanding Product Attributes Matters

- Product development
  - Over 25% of firms' revenues and profits comes from new products (McKinsey 2017)
  - Discovering and measuring key product attributes are critical to good product development
- Extracting product attributes that matter
  - Traditional methods are expensive and time-consuming
  - Increasing availability of rich unstructured (reviews) and structured (ratings) data
  - But public data contain biases
  - We need advanced techniques to control for these biases

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Gaps in Relevant Literature

- Challenges in understanding latent representations and design (Locatello et al. 2019; Burnap et al. 2022; Dew et al. 2022; Sisodia et al. working paper)
- Biases in public reputation system (Li and Hitt 2008; Moe and Schweidel 2012; Mayzlin et al. 2014; He et al. 2022; Chakraborty et al. 2022)

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Research Questions and Challenges

- How can we extract and accurately measure product attributes from structured and unstructured data on consumer evaluation of products?
  - Simultaneously extracting and measuring product attributes
  - Controlling for poor quality of data
- Which extracted product attributes contribute most to sales?
  - Modeling and interpreting the relationships between extracted unobservable/implicit attributes and sales

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Our Contributions

- To the literature on new product development: A general model to extract unobservable attributes from publicly available structured and unstructured data
- To the literature on ML applications in marketing: A theory-driven model that overcomes bias and enhances explainability while recovering and identifying sales-driving product attributes

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Theory

Recovering the True Value with Biased Measures

We start with Theorem (Hu and Schennach 2008): Given three measures, X_1, X_2, and X_3 of the true value $X^\star$, if the three measures are correlated through only the true value (conditional independence restriction, CI restriction hereafter), i.e., if

$$f_{X_1X_2X_3}(x_1, x_2, x_3) = \int_{\mathcal{X}^*} f_{X_1|X^*}(x_1 \mid x^*) f_{X_2|X^*}(x_2 \mid x^*) f_{X_3X^*}(x_3, x^*) dx^*$$

for all $x_1 \in \mathcal{X}_1, x_2 \in \mathcal{X}_2, x_3 \in \mathcal{X}_3$, then $\left(f_{X_1|X^*}, f_{X_2|X^*}, f_{X_3X^*}\right)$ is uniquely identified.

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Finding Common Variations Using Eigendecomposition

Defining linear operators

$$L_{B|A}: \ \mathcal{G}(\mathcal{A}) \mapsto \mathcal{G}(\mathcal{B}) \text{ with } \left[L_{B|A}g\right](b) \equiv \int_{\mathcal{A}} f_{B|A}(b \mid a)g(a)da,$$

$$\Delta_{b;A}: \ \mathcal{G}(\mathcal{A}) \mapsto \mathcal{G}(\mathcal{A}) \text{ with } \Delta_{b;A}g \equiv f_{B|A}(b \mid \cdot)g(\cdot).$$

Then

$$L_{x_2;X_1|X_3} = L_{X_1|X^*}\Delta_{x_2;X^*}L_{X^*|X_3}, \tag{1}$$

$$L_{X^*|X_3} = L^{-1}_{X_1|X^*}L_{X_1|X_3}, \tag{2}$$

and we can use eigendecomposition to solve:

$$L_{x_2;X_1|X_3}L^{-1}_{X_1|X_3} = L_{X_1|X^*}\Delta_{x_2;X^*}L^{-1}_{X_1|X^*},$$

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Recovering the True Value Using Neural Network (NN)

- Hu et al. (2023): Using NN to search X^* that satisfy the conditional independence (CI) restriction

$$f_{X_1,X_2,...,X_N|\hat{X}^*} = f_{X_1|\hat{X}^*} f_{X_1|\hat{X}^*} \cdots f_{X_N|\hat{X}^*}$$

  where $\hat{X}^* = \mathrm{NN}(X_1, X_2, ..., X_N)$
- Limitations: Interpretability and convergence
- We propose **N**on-**L**atent **V**ariational **A**uto**E**ncoder (NLVAE)
  - NLVAE disentangles latent representations using (biased) measures, which enables the interpretation of the latent space
  - NLVAE uses the CI restriction to regularize the latent space and approximate the distributions of the true values
  - NLVAE utilizes the technique of automatic differentiation and mild assumptions to ensure identification

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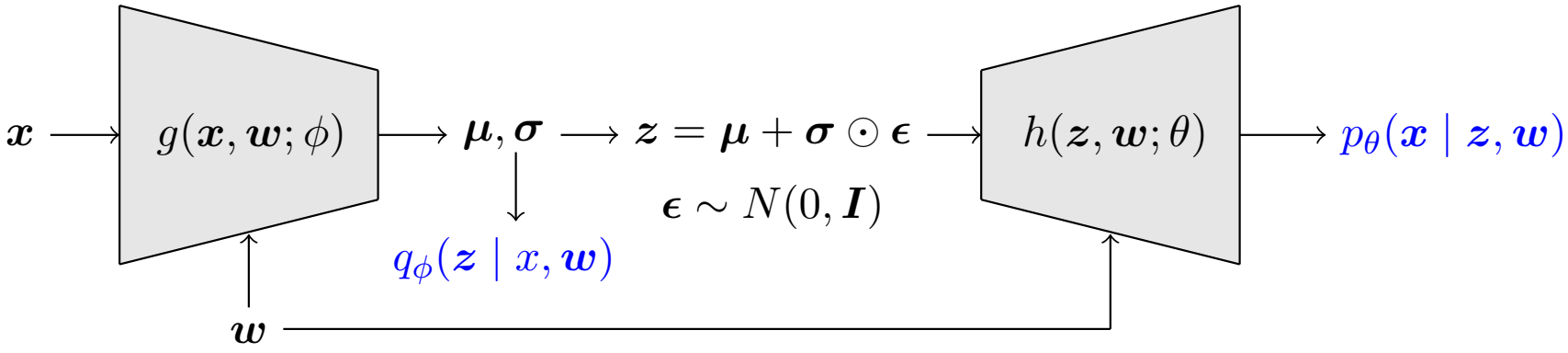
Conditional Variational Autoencoder (CVAE)

We use the architecture of CVAE (Sohn et al. 2015)

- We reconstruct only one measure that contains most information
- Other measures are treated as conditions and help decompose and regenerate the main measure



The optimization target is given by:

$$\mathcal{F}_{\mathrm{CVAE}}(\theta, \phi) = \frac{1}{N} \sum_{i=1}^{N} \left[-\log p_\theta \left(x_i \mid \boldsymbol{z}_i, \boldsymbol{w}_i \right) + \mathrm{D}_{\mathrm{KL}} \left[q_\phi \left(\boldsymbol{z}_i \mid x_i, \boldsymbol{w}_i \right) \| \Pr \left(\boldsymbol{z}_i \right) \right] \right]$$

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NLVAE = CVAE + CI Restriction

The CI restriction over the enconding space leads to a new optimization target:

$$\min \mathcal{F}_{\mathrm{CVAE}}(\theta, \phi) \ s.t. \ \mathrm{D_{KL}} \left[f_{X \boldsymbol{W}_j Z_j} || f_{X Z_j} \prod_{k=1}^{K} f_{W_{jk}|Z_j} \right] \leq \varepsilon$$

Incorporating the Karush–Kuhn–Tucker (KKT) conditions:

$$\mathcal{F}_{\mathrm{KKT}}(\theta, \phi; \beta) = \frac{1}{N} \sum_{i=1}^{N} \left[-\log p_\theta \left(x^{(i)} \mid \boldsymbol{z}^{(i)} \right) + \mathrm{D_{KL}} \left[q_\phi \left(\boldsymbol{z}^{(i)} \mid x^{(i)}, \boldsymbol{w}_1^{(i)}, ..., \boldsymbol{w}_J^{(i)} \right) \| \Pr \left(\boldsymbol{z}^{(i)} \right) \right] \right]$$
$$+ \beta \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{J} \mathrm{D_{KL}} \left[f_{X \boldsymbol{W}_j Z_j}(x^{(i)}, \boldsymbol{w}_j^{(i)}, z_j^{(i)}) \| f_{X Z_j}(x^{(i)}, z_j^{(i)}) \prod_{k=1}^{K} f_{W_{jk}|Z_j}(w_{jk}^{(i)}, z_j^{(i)}) \right]$$

The KKT multiplier β is a hyperparameter, similar to the idea of β-VAE (Higgins et al. 2017)

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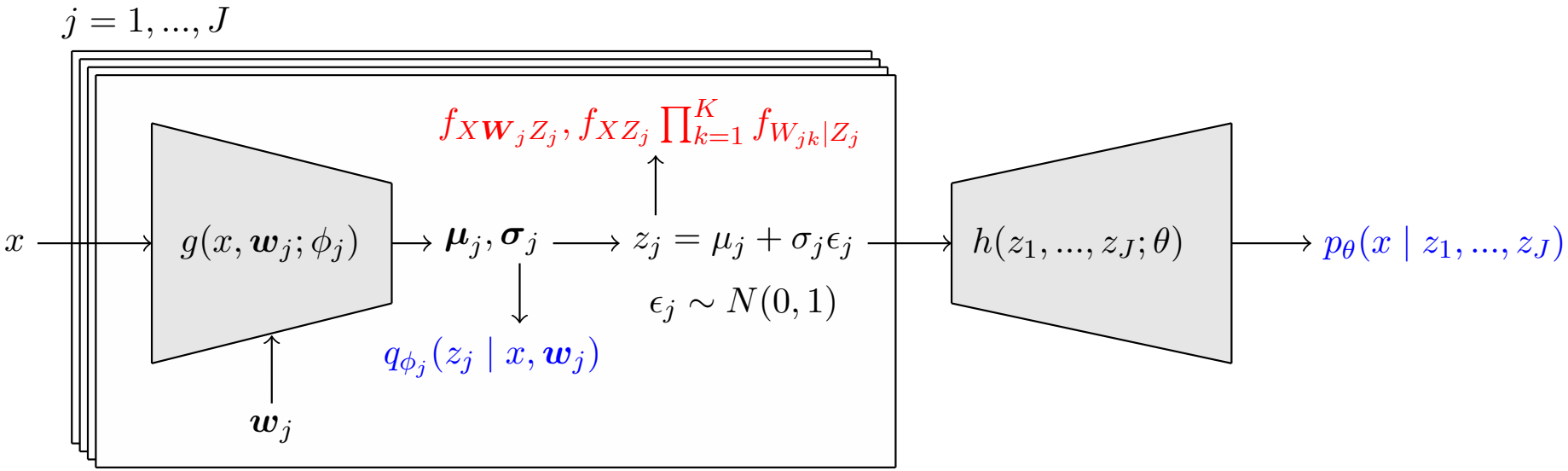
Model Architecture of NLVAE



Empirical context in our study:

- x: overall product ratings
- $\boldsymbol{w}_j = [w_{j1}, ..., w_{jK}]$: measures of the j-th product attribute
- z_j: true attribute level of the j-th product attribute

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Simulation

Hypothesized Data Generation Process

- Motivation: Since the true values are unobservable, we use a hypothesized data generation process to evaluate NLVAE
- Context and data
  - Overall product rating X_1 (the main measure)
  - Two measures (W_{2j}, W_{3j}) for each of eight attributes
  - Each observation is given by: $(x_{1i}, w_{21i}, w_{31i}, ..., w_{28i}, w_{38i})$
- Goal: Decompose true attribute levels from the overall rating
  - Recover each true attribute level z^*_{ji} from $(x_{1i}, w_{2ji}, w_{3ji})$
- Data generation process
  - $Z^*_j \sim N(0,1)$
  - $X_1 = \sum_j \delta_j Z^*_j + \varepsilon_1,\ \varepsilon_1 \sim N(0, 0.01), \sum_j \delta_j = 1$
  - $W_{2j} = (Z^*_j - 0.2)^2 + \varepsilon_2,\ \varepsilon_2 \sim N(0, 0.1)$
  - $W_{3j} = 0.02 {Z^*_j}^3 + \varepsilon_3,\ \varepsilon_3 \sim N(0, 0.1)$

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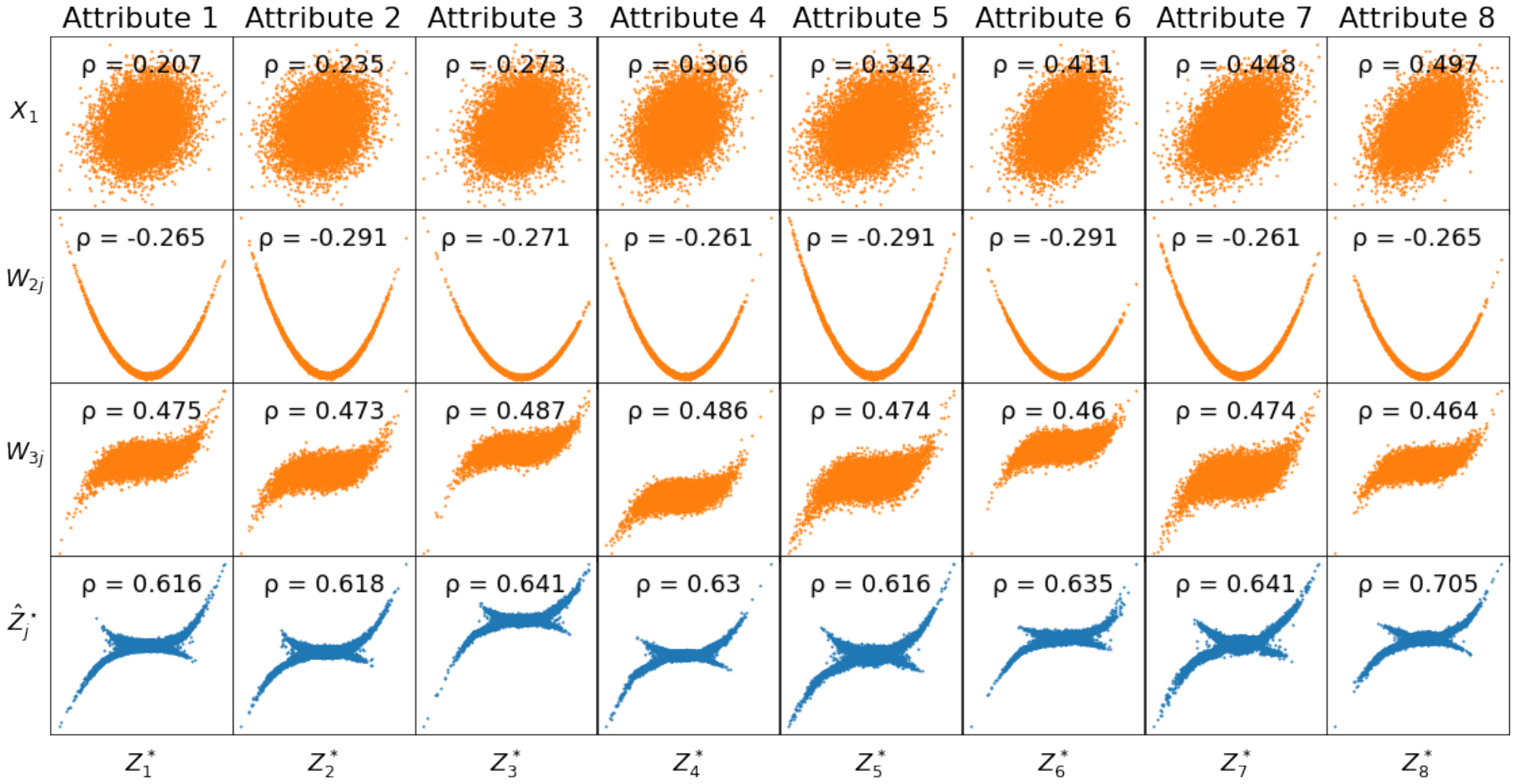
Simulation Results

While observable measures are noisy with uncertain function form, NLVAE still recovers the true attribute levels and also reduces noise.



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Empirical Application

Recovering True Attribute Levels of Video Games

- Data: Video game data from Amazon, Metacritic, and Keepa
  - $\sim$750k ratings/reviews during 2001-2023
  - Historical sales rank, price, and other relevant information
- Three measures for true attribute levels
  1. Overall product rating
  2. Topic weights of other SKUs' reviews on Amazon
  3. Topic weights of the product's reviews on Metacritic
- We use an embedded topic model (Dieng et al. 2020) to generate unnormalized topic weights
  - Good capability for processing short texts (product reviews)
  - It allows for incorporating prior knowledge
  - It allows for computing non-normalized topic weights

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Extracting 11 Attributes Using Topic Model

Attribute Name	Top 10 Words
1.Configurability	functionality, interface, configuration, configurations, interfaces, capabilities, tools, components, customization, systems
2.Technical Performance	problems, problem, fallout, troubles, shortcomings, woes, repercussions, difficulties, mistakes, inconsistency
3.Purchase Support	product, software, pricing, purchase, upgrade, products, systems, hardware, price, shipping
4.Console Compatibility	cod, downloads, gamers, gamer, console, consoles, server, gaming, bucks, videogame
5.Animation Aesthetics	visuals, soundtrack, animations, cinematic animation, textures, storytelling, soundtracks, genre, cinematography
6.Narrativeness	storyline, mythology, narrative, kind, storylines, metaphor, sort, stuff, narratives, sense
7.Family/Party Elements	mother, friends, dad, friend, father, family, daughter, grandmother, son, uncle
8.Violent Elements	guns, weapon, gun, enemy, ammo, rifle, rifles, snipers, pistol, bullets
9.Racing Elements	cars, car, racing, bike, drivers, racers, race, bikes, wheel, driver
10.Mythical Elements	stuff, kind, monsters, ones, enemy, sort, elements, guy, weaponry, creatures
11.Adventure Elements	quests, monsters, adventures, puzzles, dragons, adventure, villains, creatures, dungeon, zombies

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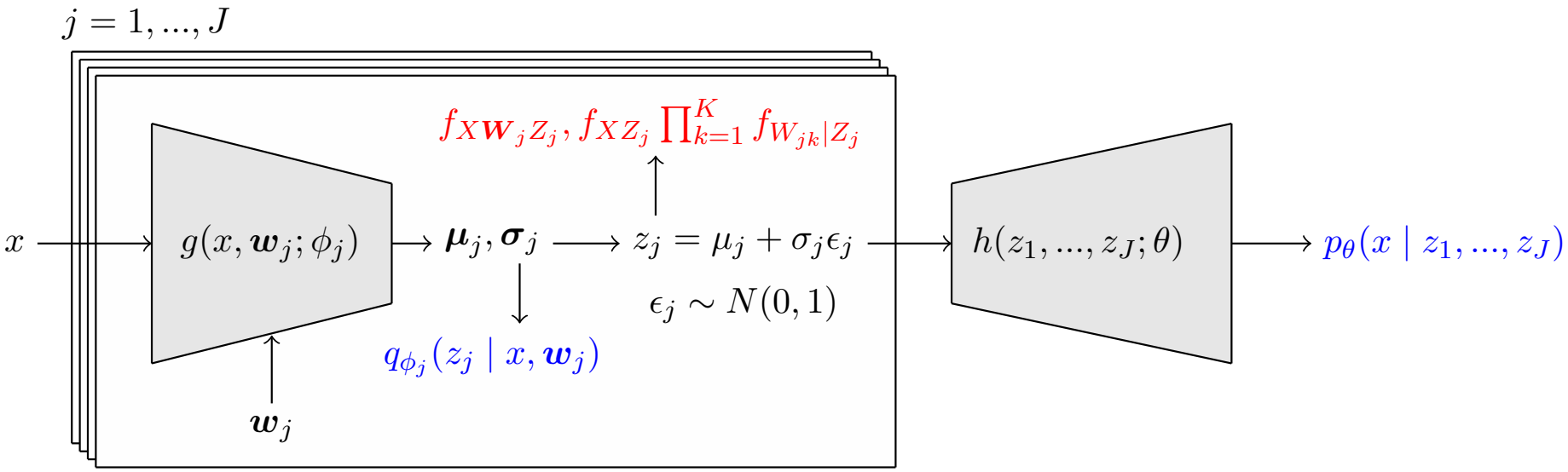
Estimating True Attribute Levels Using NLVAE



- x: overall product ratings
- $\boldsymbol{w}_j = [w_{j1}, w_{j2}]$:
  - w_{j1}: topic weights of other SKUs' reviews on Amazon
  - w_{j2}: topic weights of the product's reviews on Metacritic
- z_j: true attribute level of the j-th product attribute
  - After training, we use the encoder part only to generate estimations for the true attribute levels

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Relating Product Attributes and Sales

$$\log(\text{AmazonSalesRank}_{it}) = \beta\widehat{\text{AttributeLevel}}_{it} + \boldsymbol{\gamma}\text{Covariates}_{it} + \varepsilon_{it}$$

- Consistent with Chevalier and Mayzlin (2006) and Archak et al. (2011)

Attribute Names	Coef. with Sales Ranks	Coef. with Overall Ratings
1. Configurability	0.001	0.032**
2. Technical Performance	−0.008***	0.008
3. Purchase Support	0.026***	0.936***
4. Console Compatibility	0.003	0.058***
5. Animation Aesthetics	0.016***	0.923***
6. Narrativeness	−0.002	0.064***
7. Family/Party Elements	−0.003*	0.054***
8. Violent Elements	0.001	0.025*
9. Racing Elements	0.012***	0.948***
10. Mythical Elements	0.018***	0.957***
11. Adventure Elements	0.010***	0.889***

* $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$

- Attributes that contribute to ratings might not contribute to sales
- Some attributes may generate good feedback but do not increase (and sometimes even hurt) sales.

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Conclusion

Research Questions and Results

- How can we extract and accurately measure product attributes from structured and unstructured data on consumer evaluation of products?
  - We propose the NLVAE to extract and measure product attribute levels from online product overall ratings and reviews
  - NLVAE overcomes many types of biases in public data
  - NLVAE uses a theory-driven and interpretable model
- Which extracted product attributes contribute most to sales?
  - In the video game data we analyze, purchase support, animation aesthetics, racing elements, mythical elements, and adventure elements are positively related to sales
  - Some attributes (configurability, narrativeness, violent elements, and family/party elements) affect product ratings but do not contribute to sales

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Theoretical and Managerial Implications

- Theoretical implications
  - Important product attributes can be identified and recovered from publicly available unstructured (reviews) and structured (ratings) data
  - Attributes that influence overall rating do not necessarily contribute to sales
  - Bias mitigation and explainability of ML models can be enhanced through our theory-driven model, namely, NLVAE
- Managerial implications
  - By using NLVAE, managers can obtain unique and valuable insights beyond customer feedback
  - Managers can use publicly available reviews, ratings, and rankings data to extract sales-related product attributes and make important product improvement or new product decisions

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Thank you!

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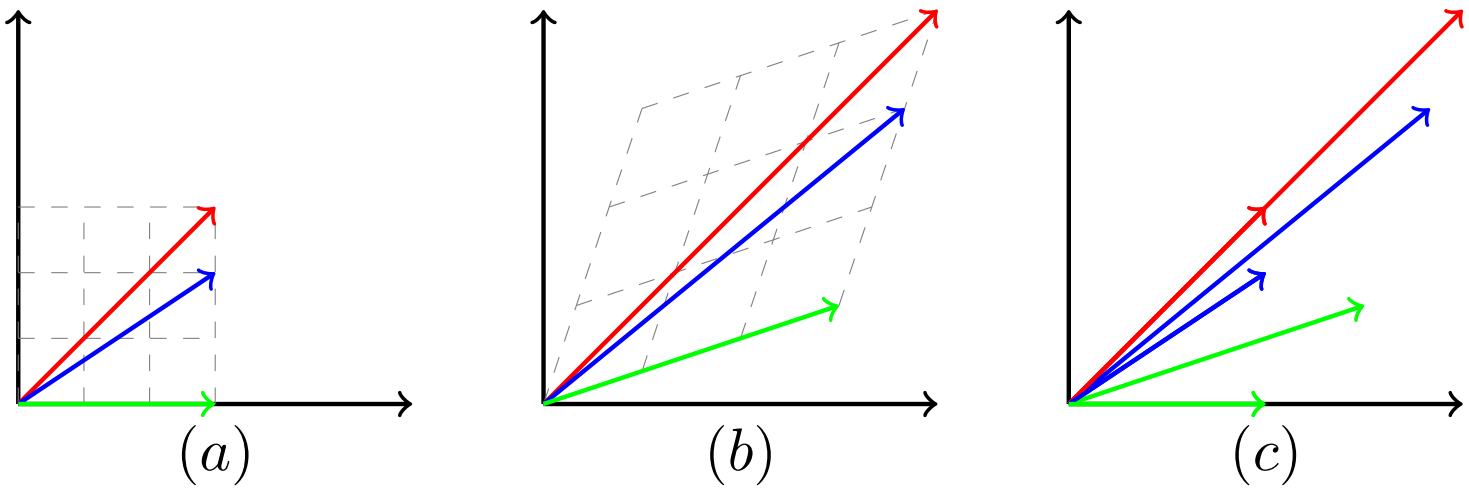
Appendix

Geometric Interpretation of Eigenvectors

$$L_{x_2;X_1|X_3} L^{-1}_{X_1|X_3} = L_{X_1|X^*} \Delta_{x_2;X^*} L^{-1}_{X_1|X^*},$$

$L_{X_1|X^*}$ is exactly the set of directions (vectors) that are fixed in the action of $L_{x_2;X_1|X_3} L^{-1}_{X_1|X_3}$



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Training Tricks

- The loss function of NN is non-convex. The model might get stuck in suboptimal points and generate unstable results
- The following procedure increases the robustness of our model
  - Step 0: train the whole model
  - Step 1: initialize the decoder and only one of encoders; keep other weights fixed
  - Step 2: retrain the model
    - If the new model yields a better KL divergence of the conditional independence restriction, use the new weights
    - Otherwise, use the old weights
    - Repeat Step 1-2 multiple times
  - Step 3: train the whole model until converge

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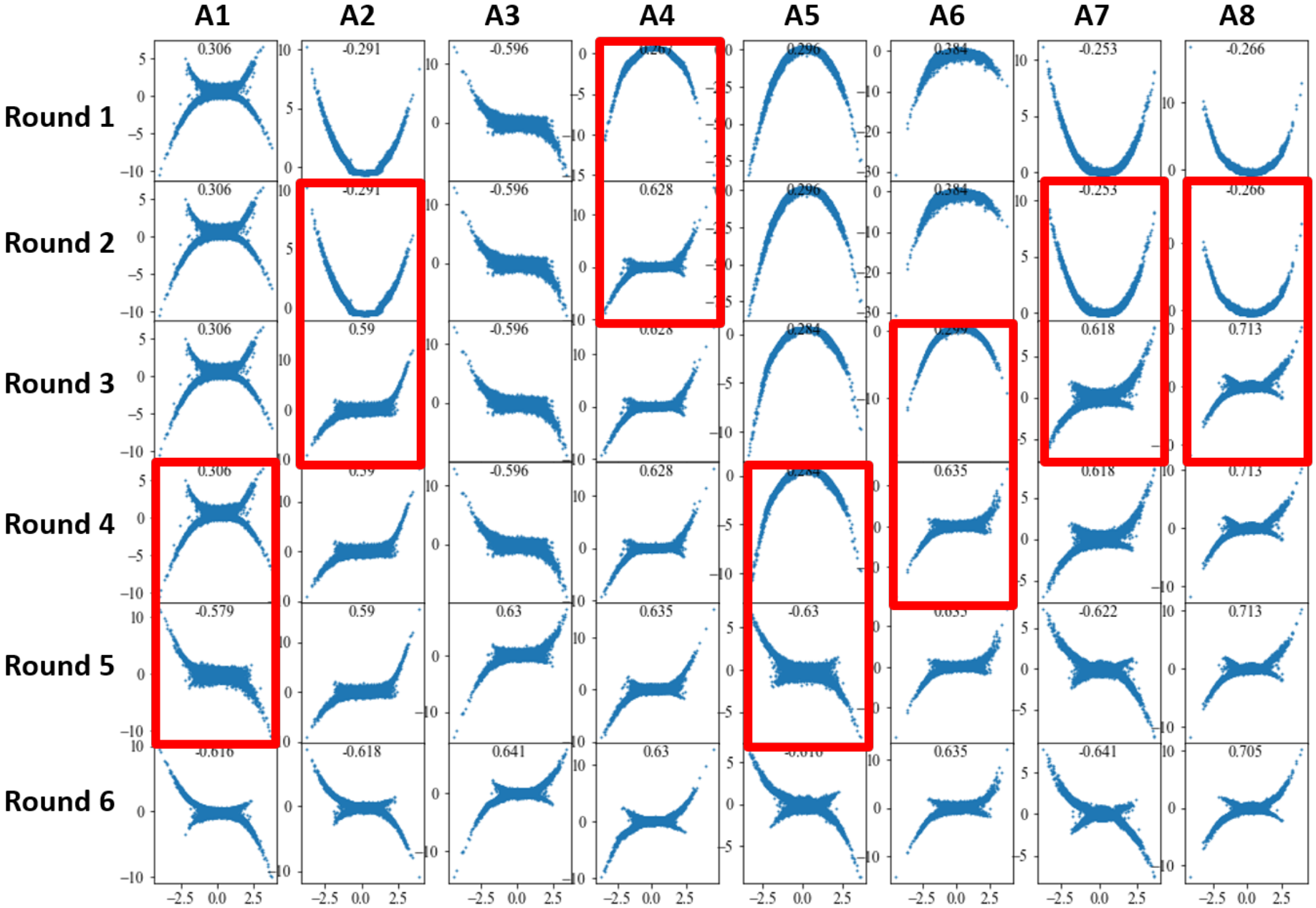
Training Tricks: Visualizing the Training Process



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Identification Tricks

- Challenge 1: the orders/directions of the true value
  - We impose a reasonable assumption that, given other attribute levels fixed, a better attribute level (s_i) leads to a higher overall rating (r):

$$P(\frac{\partial r}{\partial s_i} \leq 0) < P(\frac{\partial r}{\partial s_i} > 0)$$

  - Using automatic differentiation we can bootstrap the distribution of $\frac{\partial r}{\partial s_i}$ and determine the direction.
- Challenge 2: the scale of the true value
  - The regularization term $\mathrm{D}_{\mathrm{KL}}\left[q_\phi \| \mathrm{p}_z\right]$ helps restrict the scale
  - Cannot fully pin it down but enough for downstream tasks

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Topic Modeling in Embedding Spaces

- Data generation process
  1. Draw topic proportions $\theta_d \sim \mathcal{LN}(0, I)$ for document d
  2. For each word n in the document:
     - Draw topic assignment $z_{dn} \sim \mathrm{Cat}\left(\theta_d\right)$.
     - Draw the word $w_{dn} \sim \mathrm{softmax}\left(\rho^\top \alpha_{z_{dn}}\right)$.
- Dieng et al. (2020) compute topic weights by SoftMax$(\hat{\theta}_d)$. But in this study, we use $\hat{\theta}_d$ (unnormalized topic weights) as measures of attribute levels.
  - Independence
  - Complexity of distributions

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